

# Thermal Conductivity Measurements of Ultra-Thin Single Crystal Silicon Layers

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*Self-heating in deep submicron transistors (e.g., silicon-on-insulator and strained-Si) and thermal engineering of many nanoscale devices such as nanocalorimeters and high-density thermomechanical data storage are strongly influenced by thermal conduction in ultra-thin silicon layers. The lateral thermal conductivity of single-crystal silicon layers of thicknesses 20 and 100 nm at temperatures between 30 and 450 K are measured using joule heating and electrical-resistance thermometry in suspended microfabricated structures. In general, a large reduction in thermal conductivity resulting from phonon-boundary scattering is observed. Thermal conductivity of the 20 nm thick silicon layer at room temperature is nearly  $22 \text{ W m}^{-1} \text{ K}^{-1}$ , compared to the bulk value,  $148 \text{ W m}^{-1} \text{ K}^{-1}$ . The predictions of the classical thermal conductivity theory that accounts for the reduced phonon mean free paths based on a solution of the Boltzmann transport equation along a layer agrees well with the experimental results. [DOI: 10.1115/1.2130403]*

**Keywords:** phonon transport, silicon, heat transfer, measurement techniques, nanoscale, ultra-thin films

## 1 Introduction

Ultra-thin single-crystal silicon layers have been extensively used in the past decade for microfabrication of sensors, actuators, and transistors [1–4] as well as the fundamental study of electrical and thermal transport properties [5–10]. In recent years, the characteristic dimensions of these devices, in particular transistors [9–11], have been aggressively reduced to sub-100 nm and, therefore, more and more of them are made on silicon-on-insulator (SOI) wafers with device layers of thickness 10 to 100 nm. Self-heating and thermally induced reliability are pressing issues for advanced SOI, strained-silicon, strained Si-on-SiGe-on-insulator and strained silicon on insulator field effect transistors where the devices are separated from the silicon substrate by poor thermal conducting layers [9–17]. The lateral thermal conduction in thin silicon device layers somewhat reduces the maximum temperature rise in these transistors but its conductivity is significantly reduced due to the phonon-boundary scattering [7–10,14–17]. In addition, the thermal response times and sensitivities of miniaturized thermal analysis sensors [1], convection-based micromachined inclinometer [5], MicroElectro Mechanical Systems (MEMS) differential scanning nanocalorimeters [3], and microcantilevers for high-density thermomechanical data storage [4] are governed by lateral thermal conduction along the silicon layer. There have been limited experimental data available for lateral thermal conductivity of thin silicon layers of thickness near  $1 \mu\text{m}$ , over the temperature range of 10 to 300 K [7,8], and for silicon layers of thickness 78, 100, and 200 nm at room temperature [10]. More recently, thermal conductivity of silicon nanotubes with diameters of 22, 37, 56, and 115 nm have been measured in the temperature range 20 to 300 K [18]. There are no experimental data available for silicon layers of thickness in the range of 20–100 nm, in particular, above the room temperature, which are more relevant for the thermal design of the deep submicron SOI and strained-Si transistors.

In this manuscript, we provide data and phonon transport analysis that quantify the impact of phonon-boundary scattering on heat

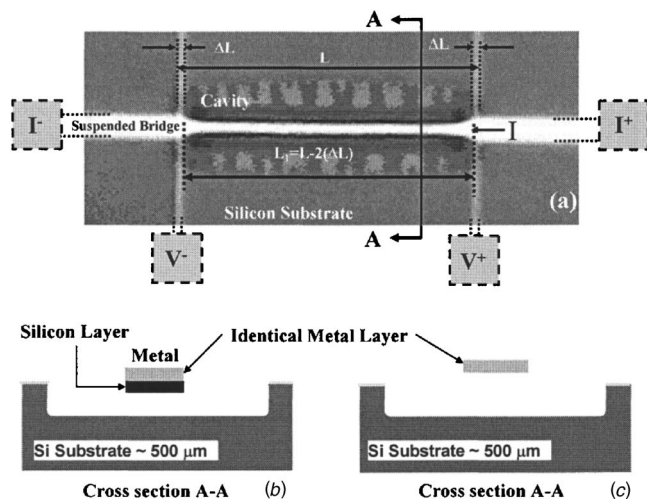
conduction in single-crystalline silicon layers of thicknesses 20 and 100 nm over the temperature range of 20–450 K. Steady-state joule heating and electrical resistance thermometry were used to measure lateral thermal conductivity of the suspended silicon/metal structures. Analyses of the heat conduction mechanisms in suspended microstructures and uncertainty in measured thermal conductivity data are presented. In addition, limitations of the above thermal conductivity measurements technique for sub-20 nm thick silicon layers are investigated and presented in the Appendix.

## 2 Experimental Structure and Procedure

The experimental data are collected using the microfabricated suspended structure shown in Fig. 1. Steady-state joule heating and electrical resistance thermometry were used to measure lateral thermal conductivity of the suspended structure. For thermal characterization of doped silicon or metal bridges, joule heating is achieved by directly passing an electric current through the bridge. However, for the nearly pure silicon layer, which is not electrically conductive, deposition of an additional metal layer is required. Clearly, the metal layer can introduce some level of uncertainty in the thermal conductivity experimental data for silicon layer hence its thermal conductivity must be measured independently. Two identical silicon-based suspended structures, with [Fig. 1(b)] and without [Fig. 1(c)] silicon layers were fabricated using nearly similar fabrication processes. Thermal conductivity of the suspended metal (CoFe and aluminum) structures are measured independently and subsequently used to extract thermal conductivity of thin silicon layers of thicknesses 20 and 100 nm, respectively.

**2.1 Microfabrication Processes.** While the final structures for thermal conductivity measurements of the 20 nm and 100 nm thick silicon layers appear to be similar (Fig. 1), the microfabrication process for these two structures are somewhat different. For the reasons that will be fully discussed in Sec. 2.3, using CoFe (instead of aluminum) layer as the heating and thermometry element is preferred. In addition, the microfabrication process described in this section was slightly modified to eliminate the impact of variation in thickness of the metal layer on the measurements of the silicon layer thermal conductivity. Hereafter,

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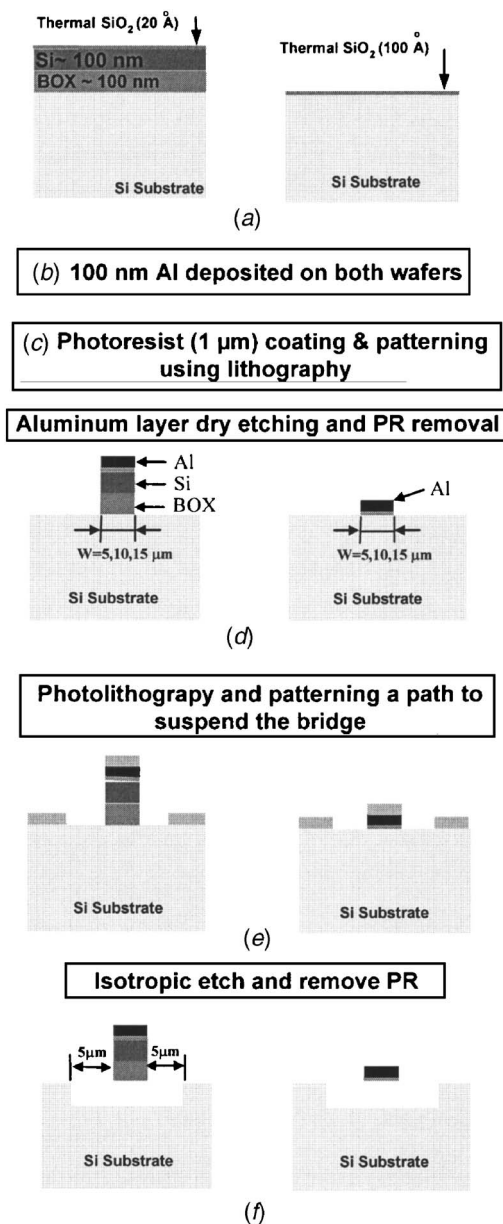


**Fig. 1** (a) The optical microscope image of the suspended structure. Cross sections of (b) the metal/silicon and (c) the metal (aluminum or CoFe) suspended structures, which are used to obtain the thermal conductivity data for silicon layer.

the aluminum/silicon and CoFe/silicon suspended structures will be referred to as AS100 and CoFeS20 structures, respectively, and the corresponding microfabrication processes will be discussed in Secs. 2.1.1 and 2.1.2, respectively.

**2.1.1 Microfabrication Process for the Aluminum/Silicon Suspended Structure (AS100).** The starting material for microfabrication of the AS100 structure is a silicon-on-insulator (SOI) wafer with a 100 nm silicon device layer and 100 nm buried silicon dioxide layer. At the first step, silicon-dioxide layers of thickness 20 Å and 100 Å are thermally grown on wafers AS100 and A100, respectively [Fig. 2(a)], followed by sputtering 100 nm thick aluminum films that will be used both as a heater and thermometer in this measurements [Fig. 2(b)]. A 1 μm thick photoresist layer (Dupont 2556) is spun onto the wafers and developed [Fig. 2(c)], followed by an ion milling process to pattern silicon/aluminum structure and aluminum bridge on the AS100 and A100 wafers, respectively [Fig. 2(d)]. A 1 μm photoresist layer (Dupont 2556) is then spun onto the wafers and developed to define the regions on the silicon substrate that would be exposed for the subsequent isotropic reactive ion etching (RIE) process [Fig. 2(e)]. The (STS) machine is then used to undercut and release the aluminum/silicon and aluminum structures [Fig. 2(f)] followed by an ion milling process using oxygen ion coupled plasma (ICP) etcher to remove the photoresist from the suspended structures.

**2.1.2 Microfabrication Process for the CoFe/Silicon Suspended Structure (CoFeS20).** The starting material for microfabrication of the CoFeS20 structure is a silicon-on-insulator (SOI) wafer with a 20 nm silicon device layer and 144 nm buried silicon dioxide layer. A 20 Å thick thermal oxide layer is grown to provide electrical insulation between the silicon and CoFe layers [Fig. 3(a)]. As we discussed previously, certain measures were adopted to reduce the uncertainty in the measurements (due to nonuniformity in deposition of metal film) that required slight modifications in the microfabrication process. At this step, the silicon overlayer is etched away from half of the 4 in. SOI wafer [Fig. 3(b)], followed by sputtering 75 nm thick CoFe layer using a cvc connexion sputtering machine [Fig. 2(d)]. A 1 μm thick photoresist layer (Dupont 2556) is spun onto the wafers and developed [Fig. 2(e)], followed by an ion milling process to pattern silicon/CoFe structure and CoFe bridge [Fig. 2(f)]. A 1 μm photoresist layer (Dupont 2556) is then spun onto the wafers and developed to define the regions on the silicon substrate that would be exposed for the subsequent isotropic reactive ion etching (RIE)



**Fig. 2** Fabrication process of 100 nm silicon film

process [Fig. 2(g)]. The STS machine is then used to undercut and release the silicon/CoFe and CoFe structures [Fig. 2(h)] followed by an ion milling process to remove the photoresist from the suspended structures.

**2.2 Experimental Setup and Procedure.** The measurements were performed in a special model Janis ST-100 continuous flow cryostat that is capable of reaching vacuum levels on the order of  $5 \times 10^{-3}$  Torr. The dies were diced and separated from identical locations in the AS100 and A100 wafers. For the case of the CoFeS20 wafer, two neighboring dies at the center of the wafer with and without the CoFe metal layer were selected and diced. While the microfabrication process usually provides nearly identical film uniformity and quality over large distances, the above additional steps were also taken for further assurances. Before discussing the experimental procedure it is important to address an important issue or concern that may come to mind regarding the microfabricated suspended structures. It is clear from Fig. 1(a) that the width of the suspended bridge varies over its length and there was initially a concern that this might have been due to

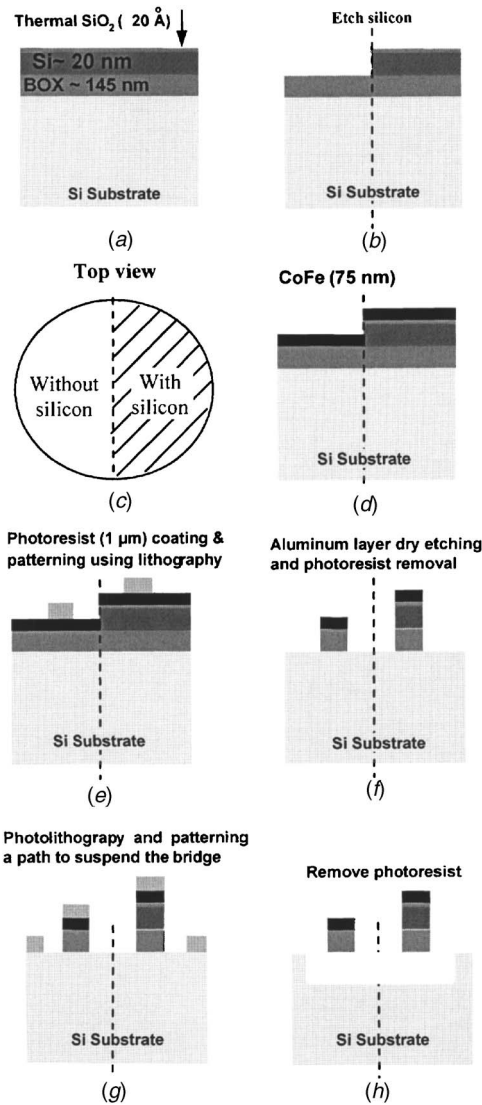


Fig. 3 Fabrication process of 20 nm silicon film

oxidization of the metal layer or the result of overetch during the surface micromachining process. This is an important factor which can directly impact the final experimental results. We resolved this matter by measuring and comparing the electrical resistivities of the suspended bridge [Fig. 2(b)] and an on-substrate section of the bridge, which shows no variations in width. The electrical resistances of the two structures were identical, which indicates the reduced dimension, as observed in Fig. 1(a), is probably due to bulging of the cantilever due to residual stress, not by oxidation/etch of the metal or silicon layer.

The dies were wire bonded in 68 pin Leadless Ceramic Chip (LCCs) chip carriers that provide electrical access to the suspended beams. The electrical resistances of the metal beam elements were measured using the four-probe measurement technique. The temperature dependency of the metal bridges electrical resistance was measured by changing the temperature of the chip carrier using a Lakeshore model 331S temperature controller (temperature stability:  $\pm 50$  mK). The accuracy of the temperature sensor used for the calibration is about  $\pm 50$  mK [19]. Steady-state joule heating and electrical resistance thermometry were used to measure the lateral thermal conductance of the suspended structures [20,21] in vacuum ( $5 \times 10^{-3}$  Torr) to minimize conduction to the air. All of the volumetric heat generated in the aluminum film flows by conduction along the composite bridge structure

toward the base and subsequently spreads out into the substrate. Therefore, heat transfer in the suspended structure may be assumed one dimensional along the structure. In addition, an analytical solution for heat conduction in the suspended bridge with uniform heat generation and surface heat losses due to radiation and molecular conduction was developed to quantitatively examine the effect of unwanted heat loss mechanisms from the bridge. Heat transfer analyses have shown that heat losses from the surface by both radiation and molecular gas conduction are negligible [9,20]. It can be shown that, assuming  $\varepsilon=1$  and for the average temperature rise in the bridge in the order of  $20^\circ\text{C}$ – $30^\circ\text{C}$  above room temperature, the heat loss due to radiation can be neglected. In the present measurements, the average temperature rise in the bridge was kept well below  $5^\circ\text{C}$ . Surface heat loss being negligible was also verified experimentally as the thermal conductivity data for different bridge lengths and widths showed very little variations. An analysis of the spreading effect into the substrate was made and concluded that the base temperature can be assumed at  $T_0$  for all practical purposes [20]. We also performed the necessary heat conduction analysis and confirmed that the portion of the beams that rest on the substrate can be assumed to have temperatures near  $T_0$ . The slight temperature rise on the order of  $0.05^\circ\text{C}$ , can be neglected compared to the average temperature rise in the beam, which is kept between  $3^\circ\text{C}$ – $5^\circ\text{C}$ . High resolution IR thermometry was also performed, which essentially verified the above analysis.

The governing equation for one-dimensional heat transfer in the metal (aluminum or CoFe) suspended bridge with negligible surface heat loss; and volumetric heat generation and the relevant boundary conditions are given as [20–22],

$$k_m(wd_m)\frac{d^2T}{dy^2} + I^2\frac{R_0}{L}[1 + \alpha_m(T - T_0)] = 0 \quad (1a)$$

$$T(y = \pm L/2) = T_0 \quad (1b)$$

where  $k_m$  is the lateral thermal conductivity of the metal film,  $\alpha_m$  is the temperature coefficient of the electrical resistance,  $T_0$  is the temperature of the substrate,  $I$  is the applied electrical current, and  $R_0$  is the electrical resistance (at  $T=T_0$ ) for the metal bridge. The dimensions  $w$ ,  $d_m$ , and  $L$  are the width, thickness, and length of the metal bridge, respectively. The above equation also assumes that the ratio of the thermal resistance across (normal to) the thickness of the bridge,  $R_{\perp,th}$ , to the thermal resistance along (parallel to) the length  $R_{\parallel,th}$ , of the bridge is on the order of  $\sim d_m/L$ , which is much smaller than unity. This condition assures that the one-dimensional heat conduction assumption is indeed valid over the entire length of the bridge except, possibly, in the regions that are away from the base, by nearly  $\sim d_m$ . Let  $\Theta = 1 + \alpha_m(T - T_0)$  in the equation and the boundary conditions, they reduce to,

$$\frac{d^2\Theta}{dy^2} + \gamma_m^2\Theta = 0 \quad (2a)$$

$$\Theta(y = \pm L/2) = 1 \quad (2b)$$

where  $\gamma_m^2 = I^2 R_0 \alpha_m / (w L d_m k_m)$ . This equation can be solved to yield

$$\Theta(y) = \cos(\gamma_m y) / \cos(\gamma_m L/2) \quad (3a)$$

or

$$T(y) = T_0 - \frac{1}{\alpha_m} \left( 1 - \frac{\cos(\gamma_m y)}{\cos(\gamma_m L/2)} \right) \quad (3b)$$

Integrating the above equation from  $-L/2$  to  $+L/2$ , the average temperature of the bridge is given by

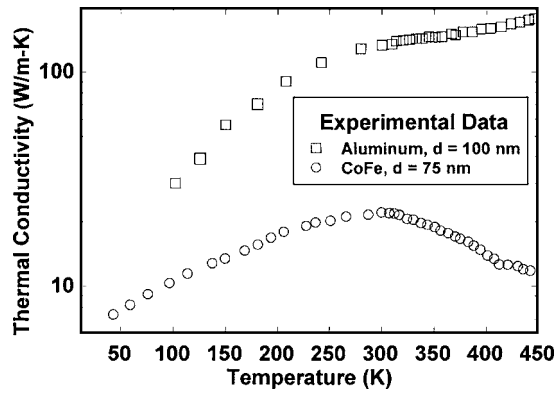


Fig. 4 Temperature dependent thermal conductivity data of the 100 nm thick aluminum and 75 nm CoFe layers as a function of temperature, which are subsequently used to obtain the thermal conductivity data for 100 nm and 20 nm thick silicon layers, respectively. The uncertainties in the data are less than 5%, which is on the order of size of the “symbols.” As a result, the uncertainty bars are not shown in this plot.

$$\bar{T} = T_0 - \frac{1}{\alpha_m} [1 - (2/\gamma_m L) \tan(\gamma_m L/2)] \quad (4)$$

The average electrical resistance of the bridge under bias is, therefore, given by

$$\bar{R}_m = R_{0,m} [(2/\gamma_m L) \tan(\gamma_m L/2)] \quad (5)$$

The lateral thermal conductivity of the metal  $k_m$  can be obtained by the measured data using Eq. (5) for the electrical resistance.

There is also a slight underetch at the bases of the suspended beams  $\Delta L \sim 4 \mu\text{m} - 10 \mu\text{m}$ , which can cause some ambiguity in defining the thermal boundary conditions described by Eq. (1b) for the governing heat conduction equation given by Eq. (1a). We attempted to account for this via evaluating the average temperature of the bridge by integrating Eq. (3) from  $(-L/2 + \Delta L)$  to  $(+L/2 - \Delta L)$ . This represents the length of the aluminum suspended bridge confined between the points of the voltage measurement Fig. 1(a).

The governing equation for one-dimensional heat transfer in the combined metal silicon [Fig. 1(b)] suspended bridge with volumetric heat generation in the metal film is very similar to Eq. (1a) and is given by

$$(k_m d_m + k_s d_s) w \frac{d^2 T}{dy^2} + I^2 \frac{R_0}{L} [1 + \alpha_m (T - T_0)] = 0, \quad (6)$$

where  $d_s$  and  $k_s$  are the thickness and the thermal conductivity of the silicon layer, respectively. Equations (1b) through (5) are still valid except that the parameter  $\gamma_m^2$  must now be replaced by  $\gamma_{s,m}^2 = I^2 R_0 \alpha_m / [w L (k_m d_m + k_s d_s)]$ . Therefore, the average electrical resistance of the silicon/metal bridge is given by

$$\bar{R}_{s,m} = R_{0,(s,m)} [(2/\gamma_{s,m} L) \tan(\gamma_{s,m} L/2)] \quad (7)$$

Given the measured thermal conductivity of the metal  $k_m$  the thermal conductivity of the silicon layer  $k_s$  can be obtained by comparing the measured electrical resistivity data and predictions of Eq. (7) for the electrical resistance.

A total number of three aluminum ( $w = 16 \mu\text{m}$ ,  $L = 500 \mu\text{m}$ , and  $w = 20 \mu\text{m}$ ,  $L = 300, 500 \mu\text{m}$ ) and three CoFe ( $w = 5, 10 \mu\text{m}$ ,  $L = 100 \mu\text{m}$ , and  $w = 15 \mu\text{m}$ ,  $L = 200 \mu\text{m}$ ) bridge structures were characterized. Experiments were conducted in vacuum and for a temperature range of 30 K to 450 K. Figure 4 shows the measured lateral thermal conductivities of the 100 nm thick aluminum and 75 nm thick CoFe layers. It becomes immediately clear that the thermal conductivity of the CoFe metal layer is by far smaller

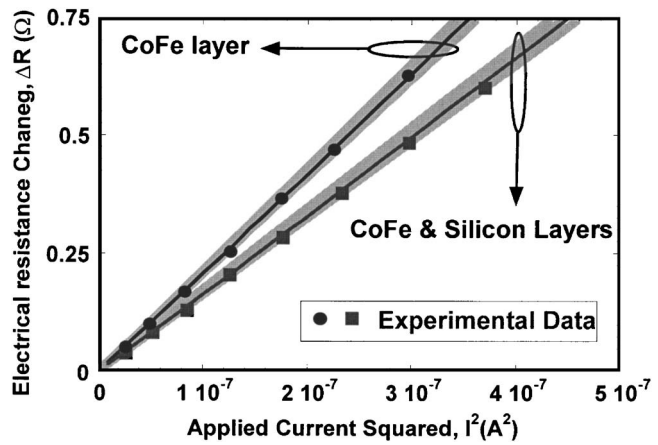


Fig. 5 Curve fits for thermal conductance along the length of the CoFe and CoFe+silicon suspended structures using Eqs. (5) and (7) to the electrical resistivity data at room temperature. The fitted thermal conductivity data for CoFe layer is subsequently used to extract the thermal conductivity of the 20 nm thick silicon layer. The uncertainties in the measured electrical resistivity data are less than 2%, which is on the order of size of “symbols.” The gray shaded areas show the  $\pm 5\%$  and  $\pm 7\%$  variations in thermal conductance of the CoFe and CoFe + silicon structures based on the predictions of the Eqs. (5) and (7).

than that of the aluminum layer and therefore more suitable for the present thermal conductivity measurements. In particular, the fact that thermal conductivity of the CoFe film is reduced at higher temperatures, compared to the aluminum layer, has a strong impact on both the feasibility of the present measurements for ultrathin silicon layers and uncertainty of the results. This matter will be further examined in the following sections. The thermal conductivity data for aluminum and CoFe metal layers are subsequently used in Eq. (7) to extract the lateral thermal conductivities of the silicon film as a function of temperature from the combined aluminum/silicon or CoFe/silicon structures. Figure 5 shows a plot of the electrical resistances of the CoFe  $\bar{R}_{\text{CoFe}}$  and CoFe/Silicon, structures as a function of the applied electrical current squared  $I^2$ . Given the thermal conductivity of the CoFe layer  $k_{\text{CoFe}}$ , the thermal conductivity of the silicon layer  $k_s$  can be obtained by comparing the measured electrical resistivity data and predictions of Eq. (7) for the electrical resistance. The applied currents were adjusted such that the average temperature rise did not exceed  $3^\circ\text{C} - 5^\circ\text{C}$ . While the increased current level would increase the sensitivity of the measurements, it would also adversely affect the measurements in three aspects. First, if the average temperature rise of the bridge exceeds  $15^\circ\text{C} - 30^\circ\text{C}$ , then the radiation loss from the suspended bridge become increasingly important. Since the emissivity of the structures has not been measured independently, then the radiation loss cannot be determined and properly accounted for. The elevated temperatures also imply that one should specifically account for both the temperature dependence of the thermal properties of the metal and silicon layers. This is due to the fact that Eq. (7) neglects the variation in thermal conductivities of the metal and silicon with temperature. For the case of CoFe/silicon structure, nonlinear behavior in the plots of  $\bar{R}_{\text{CoFe}}$  versus  $I^2$  was observed for excessive temperature rise of  $20^\circ\text{C}$ , for the measurements near room temperature. In addition, the metal bridges rapidly deteriorate at high current levels and small nonlinearity in the electrical resistivity versus temperature may cause significant error in the measurements.

**2.3 Experimental Uncertainty.** A careful consideration of the experimental uncertainty is not only important for the present measurements but also helps to understand the limitation of this

particular technique for thermal conductivity measurements of the sub-20 nm thick silicon layers. Equations (6) and (7) can be used to perform both the sensitivity and uncertainty analyses. Basically, the impact of variations in dimensions of the suspended structure Temperature Coefficient of Resistance (TCR) thickness of the silicon layer, and thickness and thermal conductivity of the metal layer on the thermal conductivity of the silicon layer can be investigated (see the Appendix). During the design of the mask, specific patterns were designed that allowed accurate measurements of the bridge widths, and lengths within  $\pm 1\%$  of  $w$  and  $\pm 0.1\%$  of  $L$ , respectively. The temperature coefficient of electrical resistances of aluminum and CoFe, is measured to be  $\sim 1.0 \times 10^{-3} \text{ K}^{-1}$  and  $2.13 \times 10^{-3} \text{ K}^{-1}$  at 300 K with the estimated error of  $\pm 2\%$ , which is due to the fact that the temperature sensor was slightly farther away from the LCC chip carrier. The measured currents and voltages are accurate within  $\pm 0.01\%$  and  $\pm 0.1\%$ , respectively. Uncertainty in the measured thermal conductivity and thickness of the metal layer are the major source of uncertainty in determination of the silicon layer thermal conductivity. Assuming that the thicknesses of the silicon and metal layers are roughly the same, the uncertainty of the measurements will be greater if the thermal conductivity of the metal layer is significantly larger than that of the silicon layer at a given temperature. For example, the thermal conductivity of the 100 nm thick aluminum layer is about three times larger than that of silicon, at room temperature. As a result, any uncertainty in thermal conductivity measurements of the metal layer (which could be the result of variations in thickness of the metal) will be magnified threefold in the final results for the thermal conductivity of silicon layer. For example, the thermal conductivity of aluminum layer increases from  $130 \text{ W m}^{-1} \text{ K}^{-1}$ , at room temperature, to  $180 \text{ W m}^{-1} \text{ K}^{-1}$ , at 450 K. In the meantime, thermal conductivity of the 100 nm thick silicon layer reduces from  $65 \text{ W m}^{-1} \text{ K}^{-1}$  to nearly  $40 \text{ W m}^{-1} \text{ K}^{-1}$  over the same temperature range. As a result, the estimated uncertainties for 100 nm thick silicon at 300 K and 450 K are 15% and 22%, respectively. The uncertainty significantly reduced ( $\sim 9\%$ ) at lower temperatures as the thermal conductivities of the aluminum dropped to  $50 \text{ W m}^{-1} \text{ K}^{-1}$ , while the silicon thermal conductivity increased to  $100 \text{ W m}^{-1} \text{ K}^{-1}$  at temperatures around 100 K.

When CoFe layer is used as a heating and thermometry element, the uncertainty is less than 10% for the entire high temperature range. This is due to the fact that the ratio of thermal conductivities of the CoFe and 20 nm thick silicon layers remains constant over temperatures from 300 K to 450 K. It is clear that the low thermal conductivity CoFe metal alloy ( $20 \text{ W m}^{-1} \text{ K}^{-1}$  at room temperature) of thickness near 75 nm can produce more reliable data for thermal conductivity measurement of the 20 nm thick silicon layer, in particular at high temperatures.

The Appendix describes a detailed sensitivity analysis that provides guidelines to minimize the contribution of the metal heater/thermometer bridge to the thermal conductivity of the thin silicon layers of thickness less than 100 nm. In general, low thermal conductivity metal alloys are excellent candidates and can significantly reduce uncertainty in the thermal conductivity measurements of thin silicon layers. In addition, the temperature coefficient of electrical resistance of CoFe was measured and found to be nearly twice of that of aluminum film, which doubled the sensitivity of the measurements. Alternative fabrication schemes and processes are under investigation to nearly eliminate the contribution of the metal layer. One way to accomplish this is by reducing the width of the metal bridge compared to the width of the suspended silicon layer.

### 3 Theory and Experimental Results

The modified form of the conductivity integral [23], which separately accounts for the contributions of transverse and longitudinal phonon modes, along with an analytical solution to the

**Table 1 The coefficients for phonon scattering rate, group velocity, and Debye temperature**

| Scattering Rate                                                                          |                        | Fitting Parameter                                        |
|------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------|
| $\tau_{\text{isotopes}}^{-1} = A_{\text{isotopes}} \omega^4$                             |                        | $A_{\text{isotopes}} = 1.32 \times 10^{-45} \text{ s}^3$ |
| $\tau_T^{-1} = B_T \omega T^4 + \tau_{\text{isotopes}}^{-1}$                             |                        | $B_T = 9.3 \times 10^{-13} \text{ K}^{-3}$               |
| $\tau_{TU}^{-1} = \frac{B_{TU} \omega^2}{\sinh(x_\omega)}, \omega_1 < \omega < \omega_2$ |                        | $B_{TU} = 5.5 \times 10^{-18} \text{ s}$                 |
| $\tau_L^{-1} = B_L \omega^2 T^3 + \tau_{\text{isotopes}}^{-1}$                           |                        | $B_L = 2 \times 10^{-24} \text{ s K}^{-3}$               |
| Group Velocity                                                                           |                        |                                                          |
| $v_T$                                                                                    | $v_{TU}$               | $v_L$                                                    |
| 5860 $\text{m s}^{-1}$                                                                   | 2000 $\text{m s}^{-1}$ | 8480 $\text{m s}^{-1}$                                   |
| Debye Temperature                                                                        |                        |                                                          |
| $\Theta_T$                                                                               | $\Theta_{TU}$          | $\Theta_L$                                               |
| 180 K                                                                                    | 210 K                  | 570 K                                                    |

Boltzmann transport equation in the thin layers is used for prediction of thermal conductivity as a function of the layer thickness and temperature [7–9]

$$k = \frac{1}{3} \sum_{j=L,T,TU} v_j^2 \int_0^{\Theta_j/T} C_{V,j}(x,T) [\tau_j(x,T) \times F(\delta)] dx \quad (8)$$

where the subscripts  $j=L, T, TU$  refer to the single longitudinal and the two transverse phonon modes, respectively,  $v_j$  is the appropriate phonon group velocity,  $\Theta_j$  is the Debye temperature of the solid,  $x = \hbar \omega / k_B T$  is the nondimensional phonon frequency,  $C_j$  is phonon specific heat per unit volume and per nondimensional phonon frequency. The Boltzmann constant is  $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$ , and Planck's constant divided by  $2\pi$  is  $\hbar = 1.602 \times 10^{-34} \text{ Js}$ . Table 1 outlines all the relevant parameters in evaluation of the thermal conductivity integral [Eq. (8)]. The boundary scattering reduction function [24]

$$F(\delta) = 1 - \frac{3}{8\delta} + \frac{3}{2\delta} \int_1^\infty \left( \frac{1}{t^3} - \frac{1}{t^5} \right) \exp(-\delta \times t) dt \quad (9)$$

depends on  $\delta = d_s / \Lambda_b$  which is the ratio of the layer thickness  $d_s$  and the appropriate phonon mean free paths for transverse or longitudinal modes  $\Lambda_b = v \times \tau_b$ . The relaxation time in the absence of phonon-boundary scattering  $\tau_b$  is what was previously determined for bulk silicon [24]. This expression assumes that phonons are diffusely scattered or emitted from the boundaries of the layer, which may slightly overestimate the conductivity reduction at temperatures below 20 K [7–9]. The fraction of phonons diffusely reflected, therefore, depends strongly on the surface roughness and the wavelength of the phonons under consideration.

The temperature-dependent thermal conductivity data and predictions of Eq. (8), as shown in Fig. 6, are in good agreement. This indicates that boundary scattering is responsible for the strong reduction in thermal conductivity, compared to the bulk silicon. The maximum thermal conductivity for the 100 nm thick silicon occurs at 100 K. As expected, the maximum thermal con-

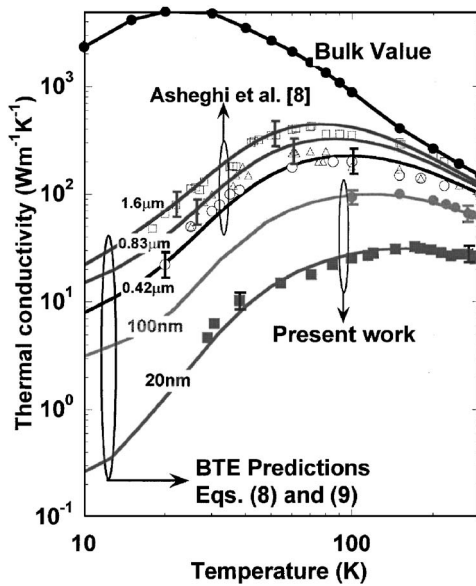


Fig. 6 The temperature dependent thermal conductivity data for the silicon layers of 20 and 100 nm thickness. The estimated uncertainties for 20 nm thick silicon layer are on the order of 12%, 8% at 30 K and 300 K. For 100 nm silicon layer, the estimated uncertainties are on the order of 9% and 15% at 100 K and 300 K, respectively.

ductivity ( $\sim 130$  K) shifts toward higher temperatures for the 20 nm thick silicon layers. The maximum thermal conductivity for the bulk silicon ( $5000 \text{ W m}^{-1} \text{ K}^{-1}$  at  $T=30$  K) separates the low-temperature region, where scattering is dominated by imperfections and surfaces, from the high-temperature region, where phonon-phonon scattering is dominant. In the ultrathin silicon layers, phonon-boundary scattering remains significant even at high temperatures, as shown in Figs. 7 and 8.

Figure 7 shows the measured lateral thermal conductivities of silicon layers as a function of thickness near room temperature. Previous [9,10] and present experimental data agree very well with the predictions, except for the silicon layer of thickness,  $d_s = 78$  nm [10]. Thermal conductivity of the 20 nm and 100 nm thin

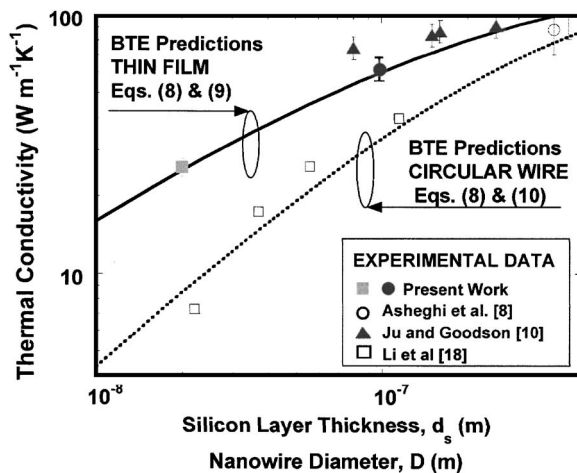


Fig. 7 Thermal conductivity of silicon layer and wires, at room temperature, as a function of thickness. Predictions based on Boltzmann transport equation (BTE) agree reasonably well with the experimental data [8,10] for thin silicon film. For nanowires, the predictions are different from the reported data [18] for  $D=22$  nm by nearly 30%.

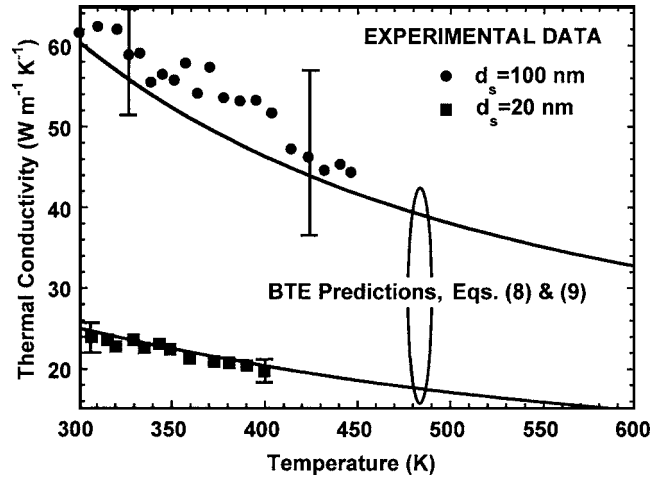


Fig. 8 Lateral thermal conductivity data and predictions for thin silicon layers at high temperatures. The estimated uncertainties for 20 nm thick silicon layer are on the order of 8%, 6% at 300 K and 400 K. For 100 nm silicon layer, the estimated uncertainties are in the order of 15% and 22% at 300 K and 450 K, respectively

silicon layers at room temperature are  $22 \text{ W m}^{-1} \text{ K}^{-1}$  and  $60 \text{ W m}^{-1} \text{ K}^{-1}$ , respectively, which are nearly 85% and 60% less than the reported bulk value of  $148 \text{ W m}^{-1} \text{ K}^{-1}$ . It is worthwhile to mention that no additional fitting parameters are used for getting agreement between the experimental data and predictions using Eqs. (8) and (9).

In addition, the graph depicts the thermal conductivity data for individual silicon nanowires of various diameters [18], which agree reasonably well (except for  $D=22$  nm) with the predictions of phonon-boundary scattering using Eq. (8) and the boundary scattering reduction function for wires with circular cross section [24]

$$F_o(\delta_o) = 1 - \frac{12}{\pi} \cdot \int_0^1 (1-t^2)^{1/2} S_4(\delta_o \times t) dt \quad (10)$$

where

$$S_n(u) = \int_1^\infty e^{-ut} (t^2 - 1)^{1/2} \cdot t^{-n} dt$$

depends on  $\delta_o = D/\Lambda_b$ , where  $D$  is the nanowire diameter.

It is suggested that the confinement of acoustic phonons and corresponding change in phonon dispersion and therefore the group velocity may lead to an increase in phonon scattering rates and, thus drastic reduction in thermal conductivity [25–27] for silicon nanowire of thickness on the order of 10 nm–20 nm. It was further claimed in Ref. [28] that for a 20 nm silicon nanowire, the overall phonon group velocity would be reduced by 40% compared with the bulk value. At the same time, changes in phonon dispersion could also modify the density of states and phonon scattering rate. For the silicon nanowire, the scattering rate for phonon-electron scattering is small compared to that of the umklapp and mass difference scattering, however, the mass-difference scattering rate is strongly increased at phonon frequency about  $5 \times 10^{13} \text{ rad/s}$  [25,28].

In order to shed light on these matters, we followed the procedure described by Bannov et al. [26] to obtain the phonon dispersion curves for 20 nm thick silicon film and nanowire of 20 nm diameter [29]. The spatial confinement in these nanostructures leads to many dispersion branches, their flattening and corresponding decrease of the phonon group velocity. It is concluded that the changes in the phonon dispersion modes for nanowire is

more pronounced compared to that of the thin film layer of comparable thickness. This may partially explain why the thin film thermal conductivity data and predictions, Eq. (8), agree well, while the predictions using Eq. (10), overestimates the nanowire thermal conductivity data by as much as 30%–40%. It is certainly worthwhile to modify the scattering rates based on the modified phonon dispersion to examine its impact on the thermal conductivity of nanowire and ultra-thin silicon layers. However, the final results will be more or less qualitative as the phonon confinement effect can also manifest itself by altering the phonon scattering rates.

The predictions of the Boltzmann Transport Equation (BTE) for silicon nanotubes (not shown here) overestimates the experimental data for a 20 nm diameter nanotube [19] at low temperatures. In contrast, the BTE predictions for a 20 nm thickness silicon layer agree well with experimental data even at 30 K. However, relatively large uncertainty in the experimental data makes it difficult to make a conclusive statement about the impact of phonon confinement effect for thin silicon layer. Perhaps, thermal conductivity measurements of 5 nm–10 nm thick layer and/or nanotubes in the temperature range of 1 K–10 K will be helpful in resolving this controversy.

#### 4 Conclusions

The lateral thermal conductivity of single-crystal silicon layers of thicknesses 20 nm and 100 nm at temperatures between 30 K and 300 K is measured, using joule heating and electrical-resistance thermometry in suspended microfabricated structures. The experimentally measured thermal conductivity and predictions based on the kinetic theory for phonon transport along thin silicon layers agree well and show significant reduction compared to the bulk value for the entire temperature range. The thermal conductivity of the 20 nm thick silicon layer, at room temperature, is nearly  $22 \text{ W m}^{-1} \text{ K}^{-1}$ , which is significantly smaller than the bulk value,  $148 \text{ W m}^{-1} \text{ K}^{-1}$ . Careful analyses of the heat conduction and uncertainty in the measured thermal conductivity data are presented. In addition, the limitations of the above technique for thermal conductivity measurements of silicon layers were carefully investigated.

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#### Appendix

The uncertainty and sensitivity analyses described here provides guidelines to minimize the contribution of the metal heater/thermometer bridge to the thermal conductivity measurements of the thin silicon layers as well as insight on the limitations of the present experimental techniques for thermal conductivity measurements of sub-100 nm thickness silicon layers.

**A.1 Uncertainty Analysis.** Metal and oxide layer's thickness and thermal conductivity variations; and nonuniformity caused by fabrication process could largely impact the uncertainty in silicon thermal conductivity data. In order to estimate the uncertainty in the results one can start with the linearized form of Eq. (7) given by

$$\Delta R = R_0 \left( \frac{L^3}{12} \cdot \frac{I^2 \rho}{w^2 d^2 k} \right) \quad (\text{A1})$$

where  $R = R_0(1 + \alpha \times \Delta T)$ . Rearranging the above equation one can arrive at the following expressions for the thermal conductivity of silicon/metal and metal structures, respectively:

$$k_{s,m} = \frac{1}{\alpha_{s,m} \Delta T_{s,m}} \left( \frac{L^3}{12} \frac{\rho I_{s,m}^2}{w^2 (d_s + d_m)^2} \right) \quad (\text{A2a})$$

$$k_m = \frac{1}{\alpha_m \Delta T_m} \left( \frac{L^3}{12} \frac{\rho I_m^2}{w^2 d_m^2} \right) \quad (\text{A2b})$$

The thermal conductivity of the silicon layer can be obtained using the following expression:

$$k_s = \frac{d_s + d_m}{d_s} k_{s,m} - \frac{d_m}{d_s} k_m \quad (\text{A3})$$

By inserting equations (A2a) and (A2b) into the above equation, one can obtain an appropriate equation for the sensitivity analysis

$$k_s = \frac{C}{\alpha} \left( \frac{C_1}{d_s + d_m} - \frac{C_2}{d_m} \right) \quad (\text{A4a})$$

where

$$C = \frac{I^2 L^3 \rho}{12 w^2 d_s}; \quad C_1 = \frac{1}{\Delta T_{s,m}}; \quad C_2 = \frac{1}{\Delta T_m} \quad (\text{A4b})$$

The above expression assumes that  $I \approx I_{s,m} \approx I_m$ ,  $\alpha \approx \alpha_s \approx \alpha_{s,m}$ , and that there are very small uncertainties associated with the measurements of  $d_s$ ,  $w$ ,  $L$ , and  $\rho$  compared to  $d_m$  and  $\alpha$ . The uncertainty in thermal conductivity of the silicon layer can be represented by [30]

$$U_{k_s} = \sqrt{\left( \frac{\partial k_s}{\partial \alpha} U_\alpha \right)^2 + \left( \frac{\partial k_s}{\partial d_m} U_{d_m} \right)^2} \quad (\text{A5})$$

Using Eq. (A4) one can arrive at the following expression for the uncertainty in thermal conductivity of silicon layer:

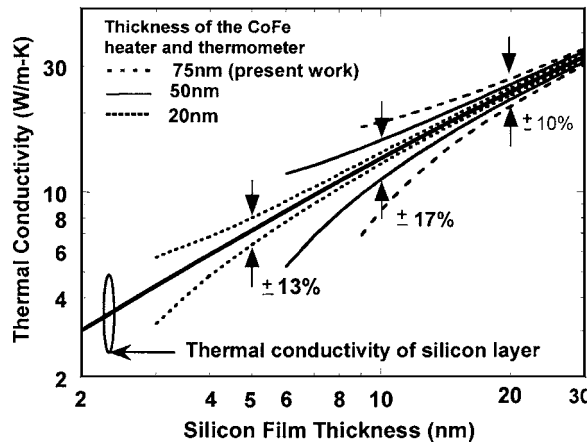
$$U_{k_s} = \frac{C}{\alpha} \times \sqrt{\left[ \left( \frac{C_2}{\alpha \cdot d_m} - \frac{C_1}{\alpha(d_s + d_m)} \right) U_\alpha \right]^2 + \left[ \left( \frac{C_2}{d_m^2} - \frac{C_1}{(d_s + d_m)^2} \right) U_{d_m} \right]^2} \quad (\text{A6})$$

The uncertainties reported in Figs. 4–7 are obtained using the above equation (see Sec. 2.3). Since  $d_m \sim 10^{-7} \text{ nm} \ll \alpha \sim 10^{-3} \text{ K}^{-1}$ , the above equation can be further simplified by ignoring the first term

$$U_{k_s} = \frac{1}{d_s} \frac{I^2 L^3 \rho}{12 w^2 \alpha} \left( \frac{C_2}{d_m^2} - \frac{C_1}{(d_s + d_m)^2} \right) U_{d_m} \quad (\text{A7})$$

It becomes clear that for a given thickness of the metal layer  $d_m$ , the uncertainty in the thermal conductivity of the silicon layer will increase if the thickness of the silicon layer is decreased. This is consistent with the trend shown in Fig. 9, which indicates that the thickness of the metal layer should be kept comparable or smaller than the thickness of the silicon layer to obtain a reasonable margin of uncertainty in the silicon thermal conductivity data.

**A.2 Sensitivity Analysis.** This section provides guidelines to minimize the contribution of the metal layer to the thermal conductivity measurement of thin silicon layer of thickness less than 100 nm. Equation (A3) can be used to investigate the impact of the variations in thickness of the metal layer on the uncertainty of the thermal conductivity data for the silicon layer. From Fig. 9, it becomes clear that 5% uncertainty in the thickness of the metal layer ( $d_m = 75 \text{ nm}$ ) results in nearly  $\pm 10\%$  uncertainty in the measurement of the thermal conductivity of 20 nm thick silicon layer



**Fig. 9** The uncertainty related to 5% variation of the metal film thickness ( $d_m=75$  nm) results in nearly  $\pm 10\%$  uncertainty in the measurement of the thermal conductivity of 20 nm thick silicon layer at room temperature. For thermal conductivity measurements of 10 nm and 5 nm thick silicon layers with uncertainty on the order of  $\sim \pm 15\%$ , metal layer thickness should be reduced to 50 nm and 20 nm, respectively.

at room temperature. For thermal conductivity measurements of 10 nm thick silicon layer with  $\pm 17\%$  uncertainty, a 50 nm thick metal layer is required. For thinner silicon layers  $d_s \sim 5$  nm, even thinner metal layer is required  $d_m \sim 20$  nm to maintain a reasonable level of uncertainty  $\sim \pm 13\%$ . While there are no fundamental limitations on thermal characterization of structures with total thickness of less than 10 nm using the present technique, the fabrication of these structures in the suspended form becomes ever increasingly more difficult. In addition, at such small dimensions accurate measurements of the thicknesses of different components of the suspended structure becomes very crucial.

In the present study, the bridges are diced from two adjacent dies, which are nearly 3 mm apart. Uncertainty in the measured thermal conductivity and thickness of the metal layer can be largely reduced (or even eliminated) if the bridges are fabricated as close as possible, preferably in the  $\sim 100$   $\mu\text{m}$  range. It becomes clear that for thermal conductivity measurements of silicon layers of thickness in the range of 5 nm–15 nm, even small variations in the dimensions and transport properties of the metal layer cannot be tolerated. Alternative fabrication schemes and processes are under investigation to nearly eliminate the contribution of the metal layer to the experimental measurements. One way to accomplish this is by reducing the width of the metal bridge compared to the width of the suspended silicon layer.

## Nomenclature

- $C$  = Phonon specific heat per unit volume,  $\text{J m}^{-3} \text{K}^{-1}$
- $D$  = Nanowire diameter, m
- $d$  = Thickness of the layer, m
- $F$  = Mean free path reduction ratio due to phonon boundary scattering for thin film, Eq. (9)
- $F_0$  = Mean free path reduction ratio due to phonon boundary scattering for circular wire, Eq. (10)
- $I$  = Applied electrical current, A
- $k$  = Thermal conductivity,  $\text{W m}^{-1} \text{K}^{-1}$
- $k_B$  = Boltzmann constant  $= 1.38 \times 10^{-23} \text{ J K}^{-1}$
- $L$  = Length of the suspended bridge, m
- $\Delta L$  = Length of the under/over etch portion of bridge, m
- $R_0$  = Electrical resistance of the metal bridge at  $T_0$ ,  $\Omega$

- $\bar{R}$  = Average electrical resistance,  $\Omega$
- $\Delta R$  = Electrical resistance change,  $\Omega$
- $T$  = Temperature, K
- $T_0$  = Temperature of the substrate, K
- $\bar{T}$  = Average temperature of the bridge, K
- $v$  = Group velocity of phonon,  $\text{m s}^{-1}$
- $w$  = Width of the suspended bridge, m
- $x$  = Nondimensional phonon frequency  $= \hbar \omega / k_B T$
- $y$  =  $y$  direction along the length of the suspended bridge, m

## Greek

- $\alpha$  = Temperature coefficient of the electrical resistance,  $\text{K}^{-1}$
- $\gamma$  = Parameter defined in equation (2a)
- $\delta$  = Dimensionless parameter for thin film  $= d_s / \Lambda_b$
- $\delta_0$  = Dimensionless parameter for circular wire  $= D / \Lambda_b$
- $\theta$  = Debye temperature, K
- $\Lambda$  = Bulk phonon mean free path, m
- $\rho$  = Electrical resistivity,  $\Omega \text{ m}$
- $\Theta$  = Temperature variable, K
- $\tau$  = Phonon relaxation time, s
- $\omega$  = Phonon angular frequency,  $\text{s}^{-1}$
- $\hbar$  = Planck's constant divided by  $2\pi = 1.055 \times 10^{-34} \text{ J s}$

## Subscript

- $b$  = Bulk silicon
- CoFe = CoFe suspended structure
- $j$  = Particular phonon mode
- $L$  = Longitudinal phonon
- $m$  = Metal suspended structure
- $m, s$  = Metal/silicon suspended structure
- $s$  = Silicon layer
- $T$  = Low frequency transverse phonons
- $TU$  = High frequency transverse phonons

## References

- [1] Furuya, M., Fujiwara, S., and Kimura, M., 2002, "Miniaturized Thermal Analysis Sensor Using Micro-Heaters on SOI Substrate," *Trans. Instit. Electrical Eng. Japan*, **122-E**, pp. 207–211.
- [2] Billat, S., Glosch, H., Kunze, M., Hedrich, F., Frech, J., Auber, J., Lang, W., Sandmaier, H., and Wimmer, W., 2001, "Convection-Based Micromachined Inclinator Using SOI Technology," *14th IEEE International Conference on MicroElectroMechanicalSystems*, Interlaken, Switzerland, pp. 159–161.
- [3] Olson, E. A., Yu, M., Efremov, E., Zhang, M., Zhang, Z., and Allen, L. H., 2003, "The Design and Operation of a MEMS Differential Scanning Nanocalorimeter for High-Speed Heat Capacity Measurements of Ultrathin Films," *J. Microelectromech. Syst.*, **12**, pp. 355–364.
- [4] King, W. P., Kenny, T. W., Goodson, K. E., Cross, G. L. W., Despont, M., Dürig, U., Rothuizen, H., Binnig, G., and Vettiger, P., 2001, "Atomic Force Microscope Cantilevers for Combined Thermomechanical Data Writing and Reading," *Appl. Phys. Lett.*, **78**, pp. 1300–1302.
- [5] Savin, A. M., Prunnila, M., Kivinen, P. P., Pekola, J. P., Ahopelto, J., and Manninen, A. J., 2001, "Efficient Electronic Cooling in Heavily Doped Silicon by Quasiparticle Tunneling," *Appl. Phys. Lett.*, **79**, pp. 1471–1473.
- [6] Pescini, L., Tilke, A., Blick, R. H., Lorenz, H., Kotthaus, J. P., Eberhardt, W., and Kern, D., 1999, "Suspending Highly Doped Silicon-on-Insulator Wires for Applications in Nanomechanics," *Nanotechnology*, **10**, pp. 418–420.
- [7] Asheghi, M., Leung, Y. K., Wong, S. S., and Goodson, K. E., 1997, "Phonon-Boundary Scattering in Thin Silicon Layers," *Appl. Phys. Lett.*, **71**, pp. 1798–1800.
- [8] Asheghi, M., Toulzaev, M. N., Goodson, K. E., Leung, Y. K., and Wong, S. S., 1998, "Temperature Dependent Thermal Conductivity of Single-Crystal Silicon Layers in SOI Substrates," *ASME J. Heat Transfer*, **120**, pp. 30–36.
- [9] Asheghi, M., Kurabayashi, K., Kasnavi, R., and Goodson, K. E., 2002, "Thermal Conduction in Doped Single-Crystal Silicon Films," *J. Appl. Phys.*, **91**, pp. 5079–5088.
- [10] Ju, Y. S., and Goodson, K. E., 1999, "Phonon Scattering in Silicon Films With Thickness of Order 100 nm," *Appl. Phys. Lett.*, **74**, pp. 3005–3007.
- [11] Rim, K., Koester, S., Hargrove, M., Chu, J., Mooney, P. M., Ott, J., Kanarsky, T., Ronsheim, P., Jeong, M., Grill, A., and Wong, H.-S. P., 2001, "Strained Si NMOSFETs for High Performance CMOS Technology," *Symposium on VLSI Technology Digest of Technical Papers*, Kyoto, Japan, pp. 59–60.

- [12] Hoyt, J. L., Nayfeh, H. M., Eguchi, S., Aberg, I., Xia, G., Drake, T., Fitzgerald, E. A., and Antoniadis, D. A., 2002, "Strained Silicon MOSFET Technology," *IEEE International Electron Devices Meeting*, San Francisco, CA, pp. 23–26.
- [13] Jenkins, K. A., and Rim, K., 2002, "Measurement of the Effect of Self-Heating in Strained-Silicon MOSFETs," *IEEE Electron Device Lett.*, **23**, pp. 360–362.
- [14] Etessam-Yazdani, K., and Asheghi, M., 2004, "Ballistic Phonon Transport on Strained Si/SiGe Nanostructures With an Application to Strained-Silicon Transistors," *IEEE ITherm Conference*, June 2004, Las Vegas, NV.
- [15] Liu, W., and Asheghi, M., 2004, "Thermal Modeling of Self-Heating in Strained-Silicon MOSFETs," *IEEE ITherm Conference*, June 2004, Las Vegas, NV.
- [16] Asheghi, M., Behkam, B., Yazdani, K., Joshi, R., and Goodson, K. E., 2002, "Thermal Conductivity Model for Thin Silicon-on-Insulator Layers at High Temperatures," Presented at the *IEEE International SOI Conference Proceedings*, Oct. 7–10, Williamsburg, VA, pp. 51–52.
- [17] Asheghi, M., Sverdrup, P., and Goodson, K. E., 1999, "Thermal Modeling of Thin-Film SOI Transistors," *IEEE International SOI Conference Proceedings*, Oct. 4–7, Rohnert Park, CA, pp. 28–29.
- [18] Li, D., Wu, Y., Kim, P., Shi, L., Yang, P., and Majumdar, A., 2003, "Thermal Conductivity of Individual Silicon Nanowires," *Appl. Phys. Lett.*, **83**, pp. 2934–2936.
- [19] F Band temperature sensor: Lake shore: <http://www.lakeshore.com/temp/sen/sdts.html> accuracy:  $\pm 20$  mK  $< 10$  K;  $\pm 55$  mK (10 K to 500 K).
- [20] Tai, Y. C., Mastrangelo, C. H., and Muller, R. S., 1988, "Thermal Conductivity of Heavily Doped Low Pressure Chemical Vapor Deposited Polycrystalline Silicon Films," *Appl. Phys. Lett.*, **63**, pp. 1442–1447.
- [21] Zhang, S., Yang, Y., Sadeghipour, M. S., and Asheghi, M., 2003, "Thermal Characterization of the 144 nm GMR Layer Using Microfabricated Suspended Structures," *ASME Summer Heat Transfer Conference*, Paper no. HT2003-40270, July 21–23, Las Vegas, Nevada.
- [22] Reifenberg, J., England, Voss, R. J., Rao, P., Schmitt, W., Yang, Y., Liu, W., Sadeghipour, S. M., and Asheghi, M., "Thermal Conductivity Measurement of Thin Aluminum Layers Using Steady-State Joule Heating and Electrical Resistance Thermometry in Suspended Bridges," *ASME International Mechanical Engineering Congress & Exposition*, IMECE-42055, November 15–21, 2003, Washington, D. C.
- [23] Holland, M. G., 1963, "Analysis of Lattice Thermal Conductivity," *Phys. Rev.*, **132**, pp. 2461–2471.
- [24] Sondheimer, E. H., 1952, "The Mean Free Path of Electrons in Metals," *Adv. Phys.*, **1**, pp. 1–42.
- [25] Nishiguchi, N., 1996, "Electron Scattering due to Confined and Extended Acoustic Phonons in a Quantum Wire," *Physica B*, **58**, pp. 1494–1497.
- [26] Bannov, N., Aristov, V., and Mitin, V., 1995, "Electron Relaxation Times due to the Deformation-Potential Interaction of Electrons With Confined Acoustic Phonons in a Free-Standing Quantum Well," *Phys. Rev. B*, **51**, pp. 9930–9942.
- [27] Zou, J., and Balandin, A., 2001, "Phonon Heat Conduction in a Semiconductor Nanowire," *J. Appl. Phys.*, **89**, pp. 2932–2938.
- [28] Balandin, A., and Wang, K., 1998, "Significant Decrease of the Lattice Thermal Conductivity due to Phonon Confinement in a Free Standing Semiconductor Quantum Well," *Phys. Rev. B*, **58**, pp. 1544–1549.
- [29] Liu, W., and Asheghi, M., 2005, "Phonon Confinement Effects in Silicon Film and Nanowire," in preparation.
- [30] Holman, J. P., 1984, *Experimental Methods for Engineers*, McGraw-Hill, New York, pp. 50–57.